AI OPEN
DATA CENTER

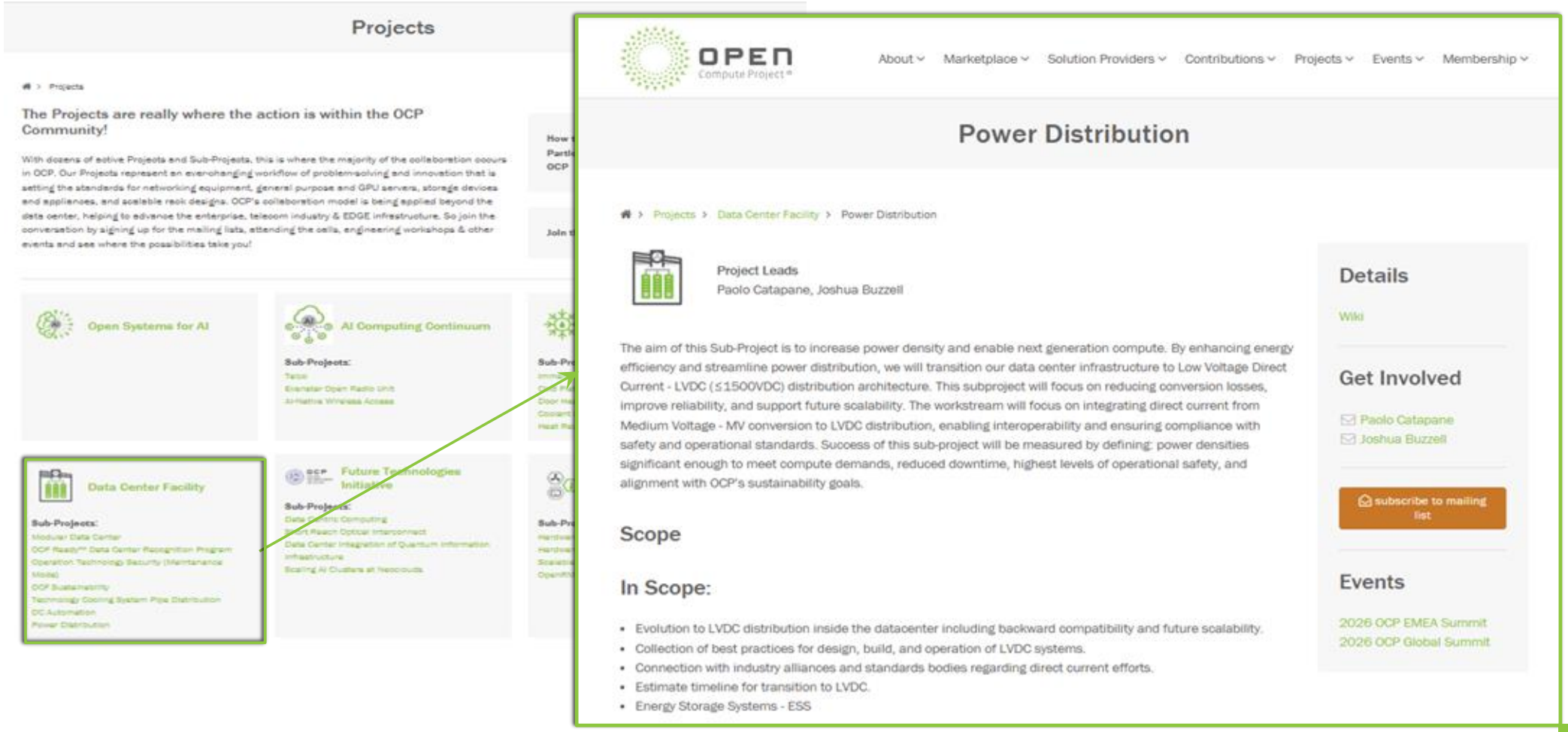
LVDC Power Distribution OCP project

Paolo Catapane *Senior Global Product Manager / ABB*

Joshua Buzzell

OCP SILVER Member

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Link to the sub-project page

OCP Power Distribution sub-project

LVDC Evolution and Scalability

Focus on evolving low voltage direct current distribution within datacenters with backward compatibility and scalability.

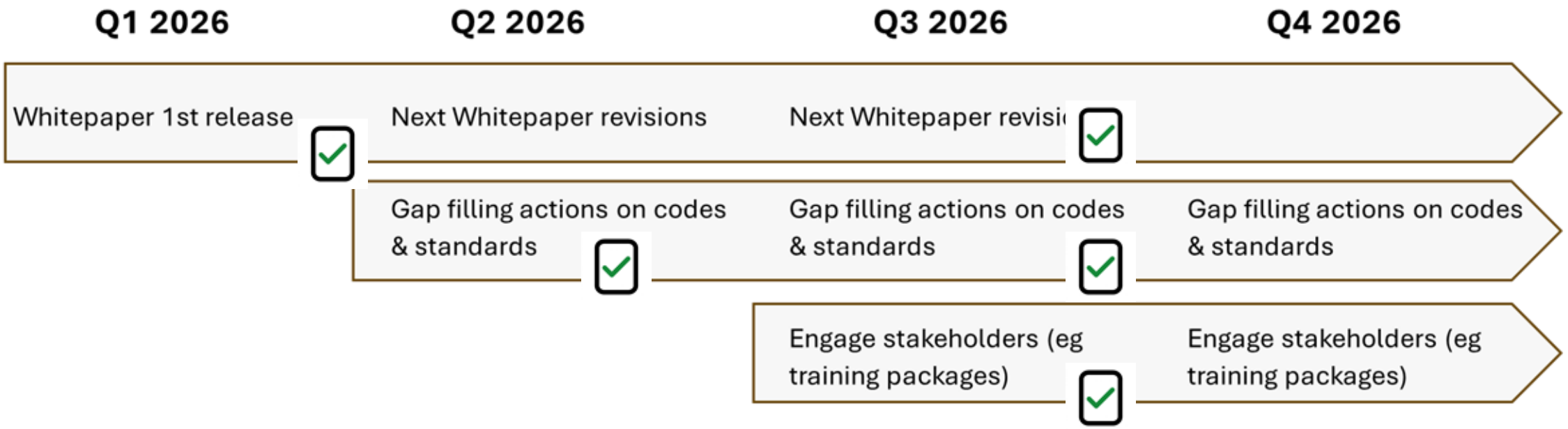
Best Practices in LVDC Systems

Compile best practices for design, construction, and operation of LVDC power systems to ensure efficiency.

Industry Collaboration

Engage with industry alliances and standards organizations to align on direct current power efforts.

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System Architecture

- JP Buzzell (Eaton)

Voltage Levels & Power Quality

- Brian Deckard (Siemens)

Polarities & System Earthing

- Laurens Mackay (DC Opportunities)

Protection and Safety

- Antonello Antoniazzi (ABB)

Main Components (high level requirements)

- Tony Landry (Schneider)

Codes and Standards

- Lisa Grams (UL Solutions)

Control & Operation

- Ke Sun (Vertiv)

EMC

- Scott Hunter (UL Solutions)

Joint Workstream DCF PD + R&P

- Rudy Wang (Delta)
- Aurelio Rodriguez Echevarria (AMD)

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800+
subscribers in groups.io

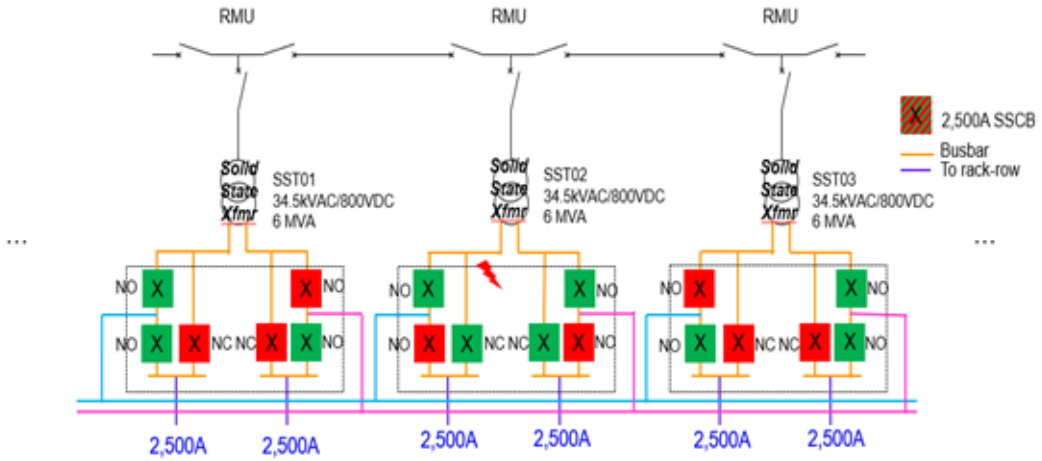
100
workstream leads + active contributors

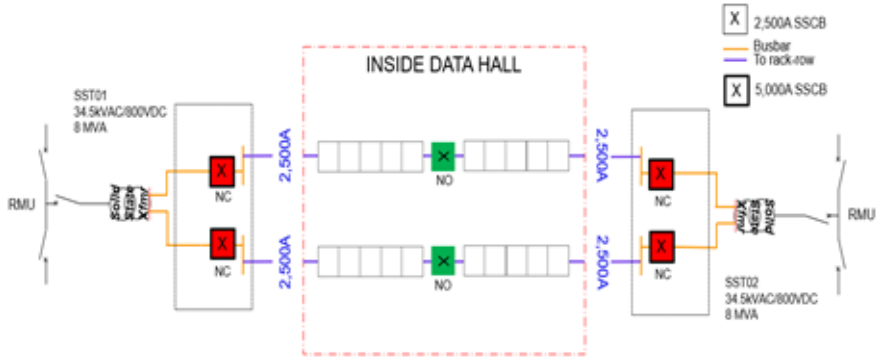
250
white paper pages

10 / 2
months / WP releases

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System architecture





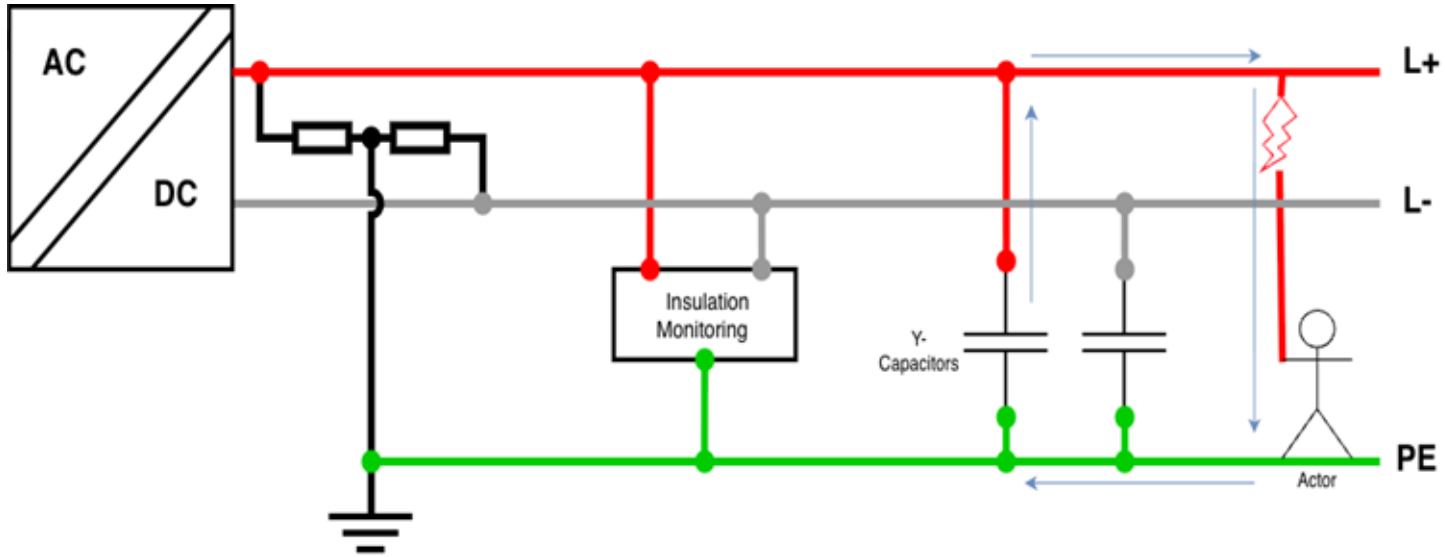
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Voltage & Power Quality

- Updated background context on DC vs. AC to the IT rack
- Updated scope consideration for energy storage circuit connection to converter
- Updated information on slash voltage rating and discussion of nominal vs. floating voltages
- Updated clearer visuals from recent OCP EMEA slides
- Added continuous voltage drop line loss impact example
- Added droop curve operation example

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Polarities & system earthing



IT earthed system where system Y-capacitance discharges into human touch fault and resulting fault current path

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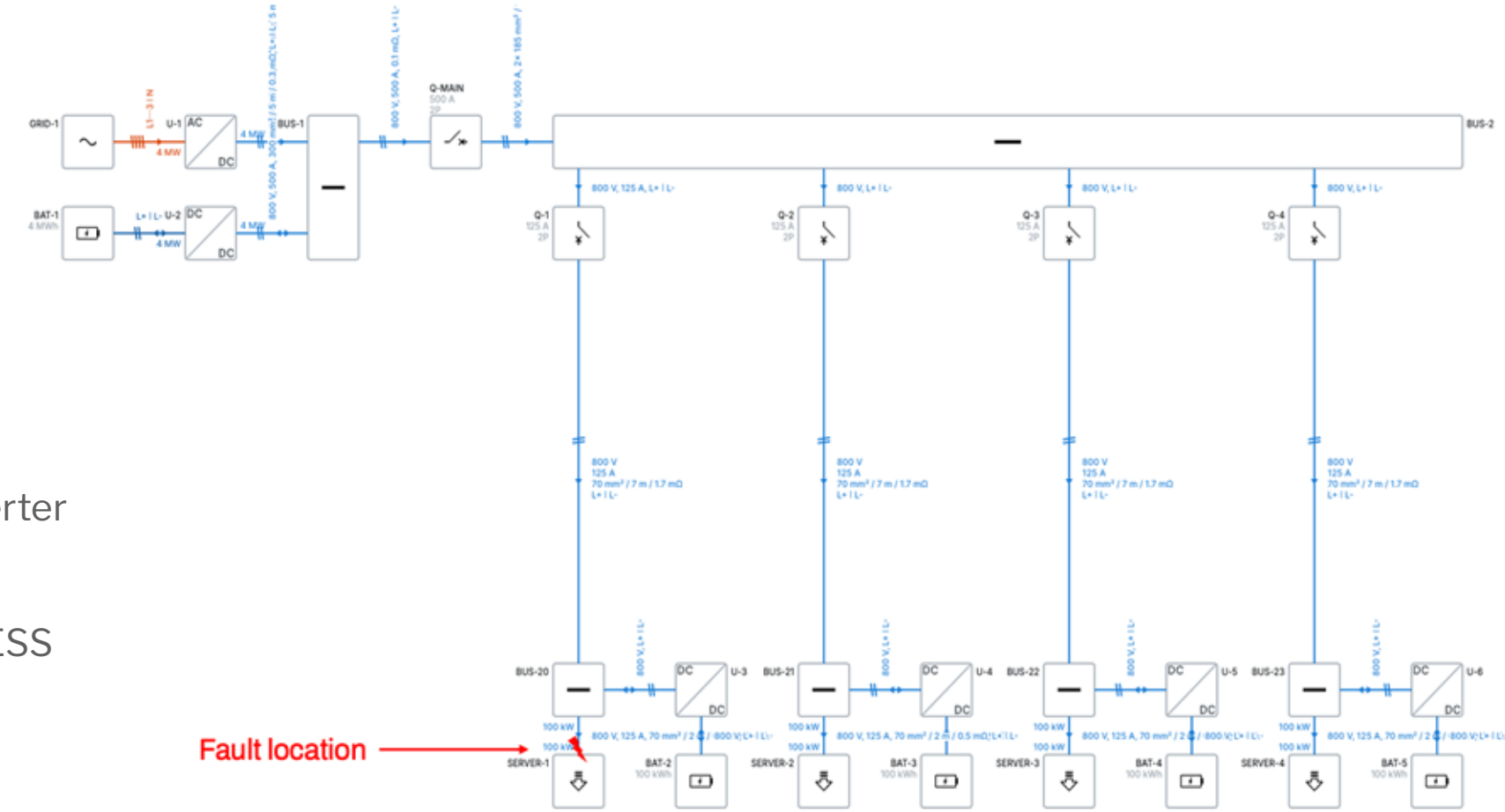
Polarities & system earthing

- IT grounded system
  - Only one active IMD
  - Safety limitation: Entire facility's Y-Capacitance could discharge through a person
  - Real-world example: EV charging Y-Capacitance limits
  - CCS at ≤ 400kW, IEC 61851-23/ISO 17409 mandates 5uF system limit
  - MCS at e.g. 1MW, Upcoming IEC 61851-23-3 with Y-Capacitance system limit of 30uF
- TN-S grounded system
  - Implementing RCDs at all branch circuits or racks, allows for larger scale systems, earth fault currents can be identified and cleared locally by the RCD.
  - Y-Capacitance discharge is limited to one circuit.

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Protection & safetv - fault analvsis

- 4 MW AC service
- 4 MW DC/DC BESS converter (600 V batteries)
- 4 x 100 kW racks
- 4 x 100 kW distributed BESS (600 V batteries)

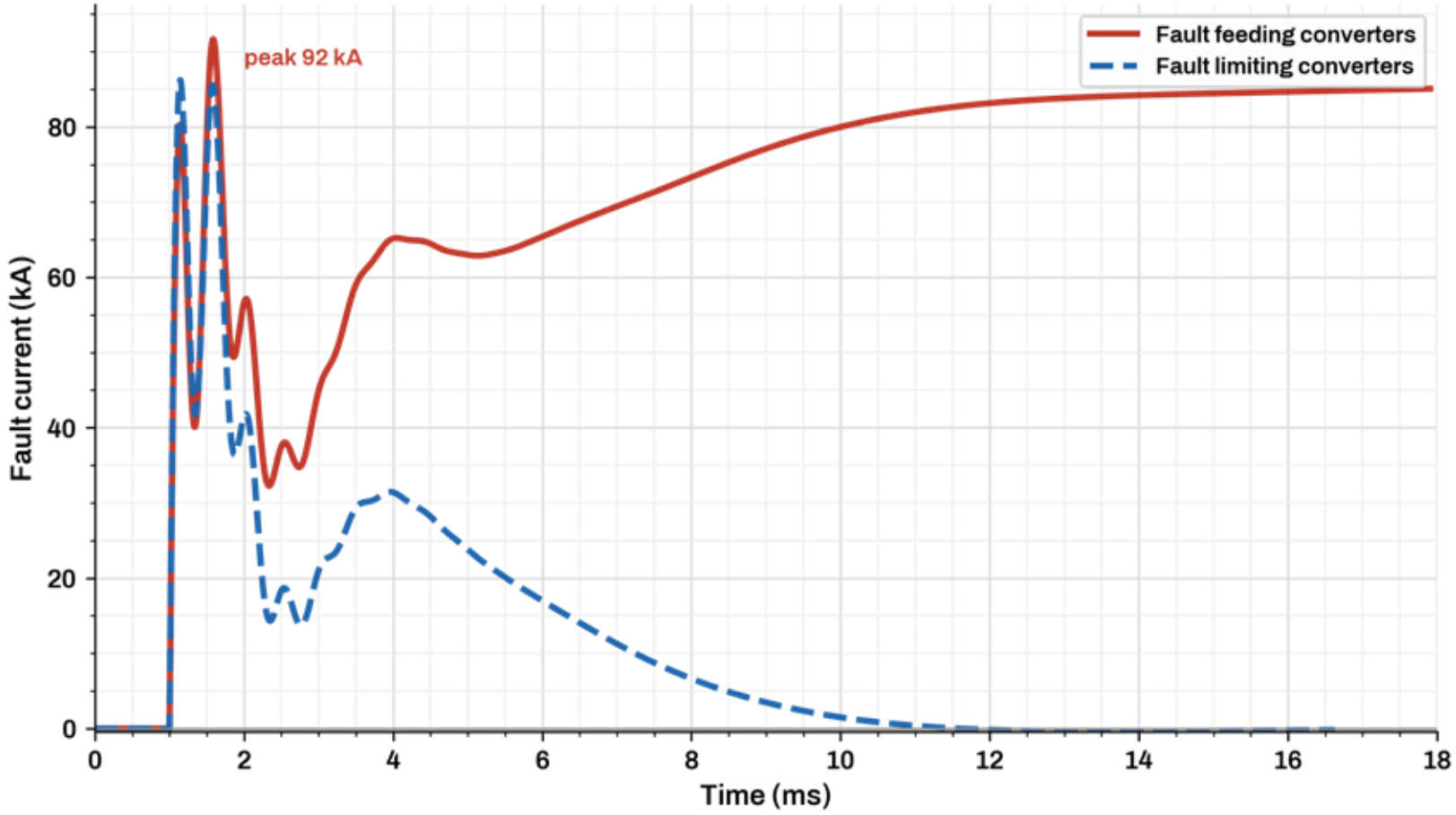


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Protection & safety - fault analysis

The steady-state current depends on the converter:

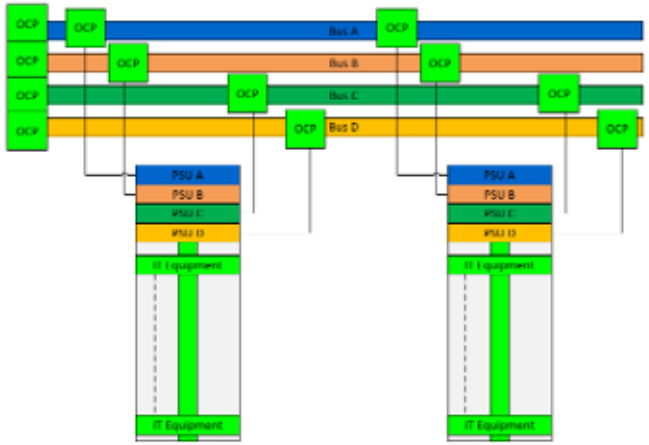
- zero (or small multiple of the nominal current) for fault-limiting converters
- large for fault-feeding converters



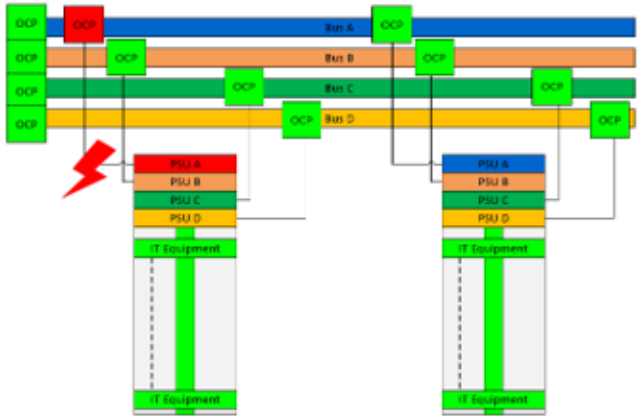
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Protection & safety - fault analysis

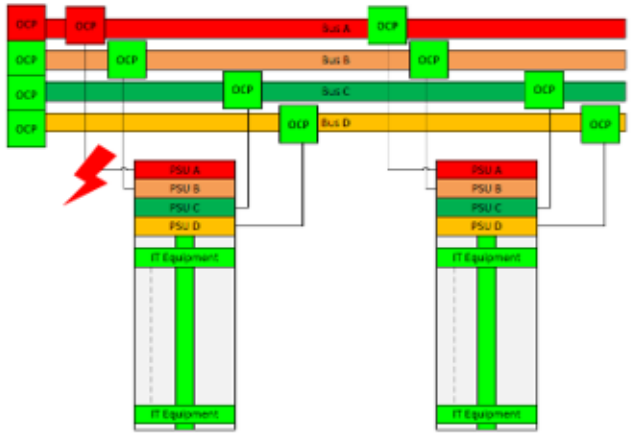
4-to-make-3 redundant architecture.



Normal operation.
Each PSU works at 75% of the rated power



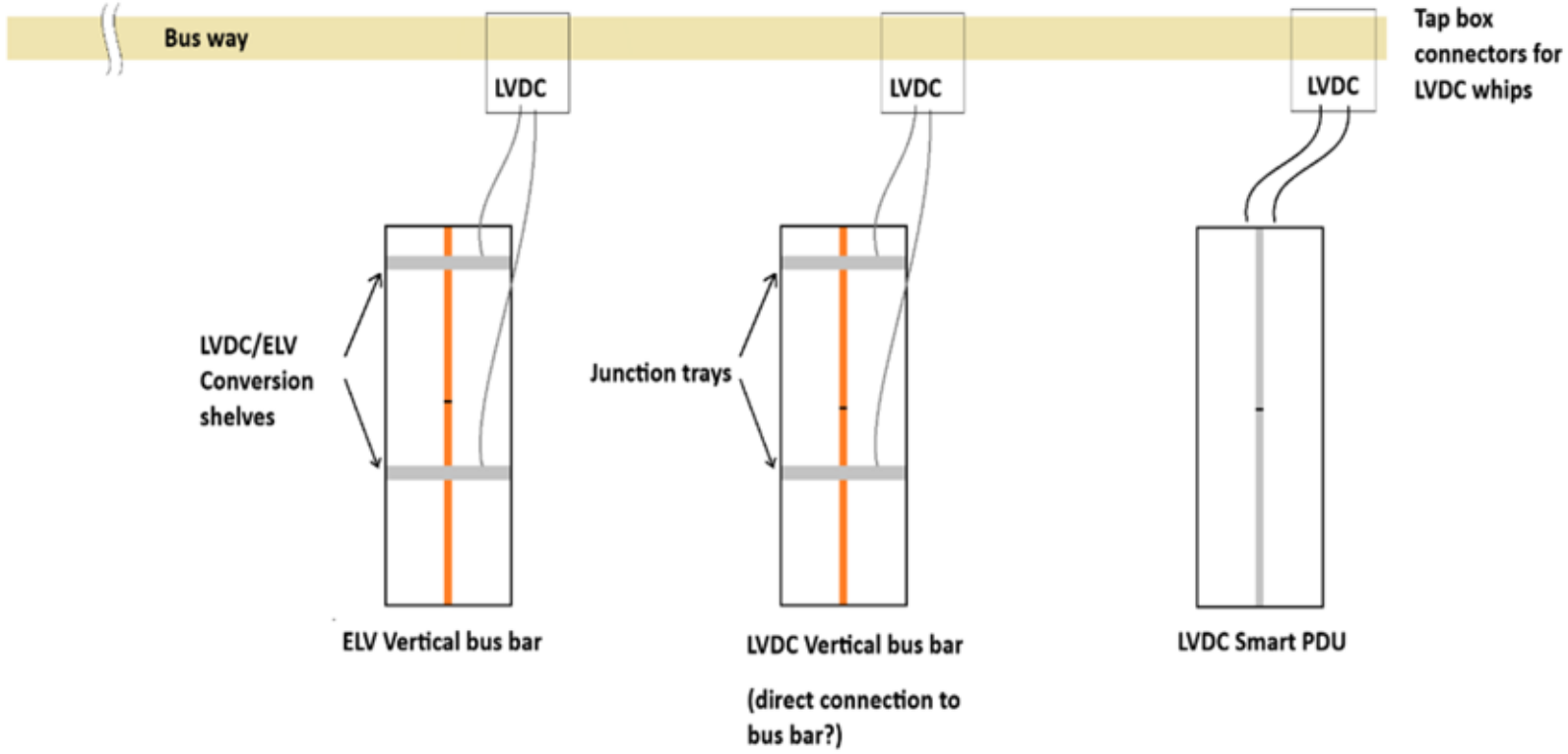
Fault with coordination of protection devices: the branch protection trips, the feeder protection does not.
On the LH rack: 3 PSUs operating at 100% of the nominal power



Fault without coordination. Both the feeder and the branch trip, but the other buses continue feeding the rack.
On both racks: 3 PSUs operating at 100% of the nominal power

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Joint Rack&Power workstream

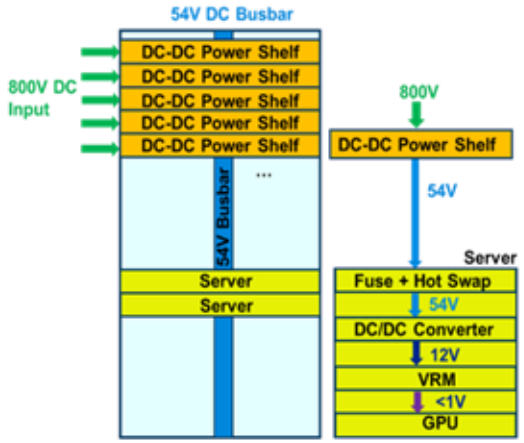


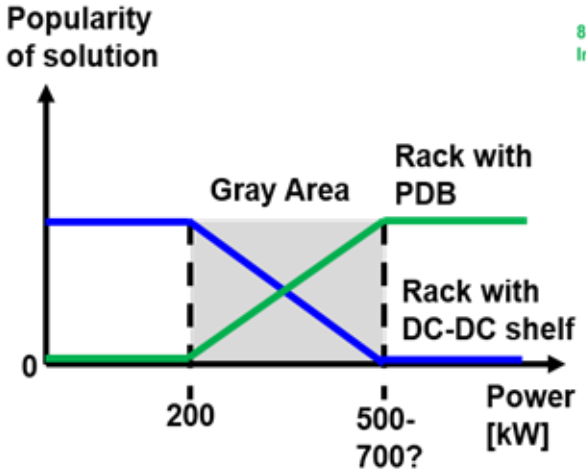
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Joint Rack&Power workstream

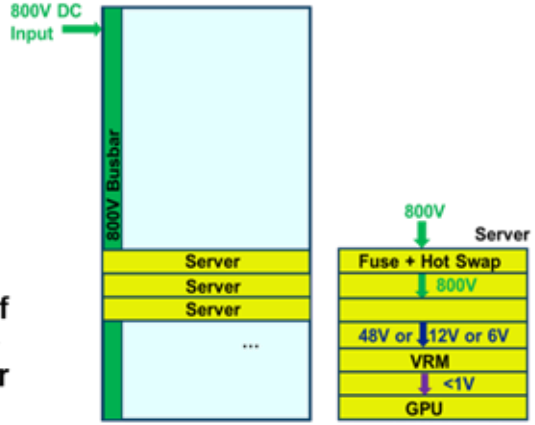
DC Power Architecture in High Power 800V IT Racks

- 800V IT Rack with dc-dc power shelf





- 800V IT Rack with power distribution board (PDB)

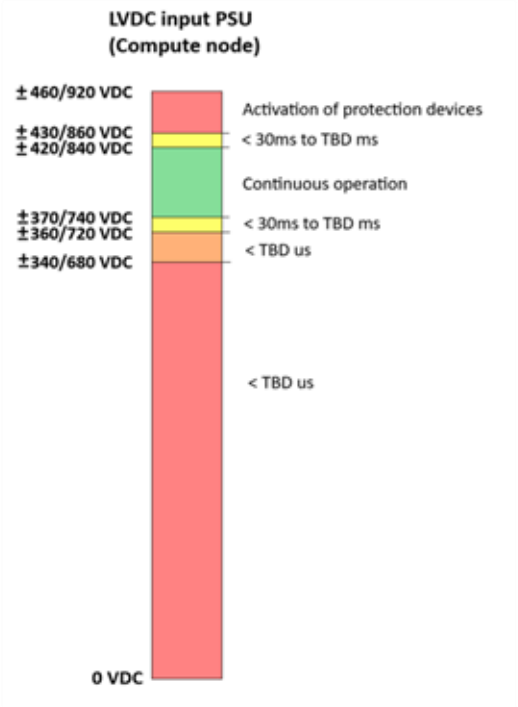


- Suitable for lower power levels because of limited current handling on 54V busbar
- Suitable for higher power levels because of 800V busbar

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Joint Rack&Power workstream

DC Distribution Voltage Range and Definition of Operation

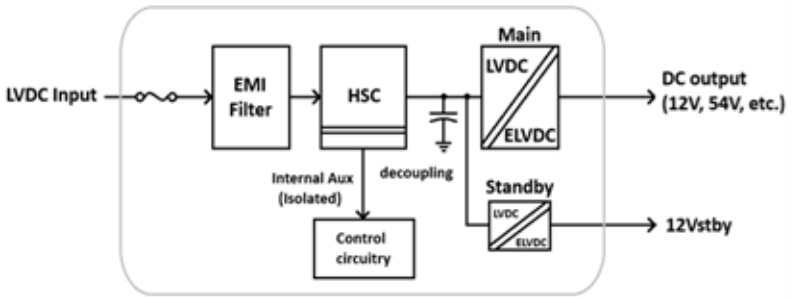


- **Continuous Operation**
  - **Power conversion can operate continuously without tripping any protection**
- **Transient 1 Operation**
  - **Power conversion can operate momentarily without tripping any protection (10s of ms)**
  - **Operation in this region allows different sources to stabilize their current share after a transient.**
- **Transient 2 Operation**
  - **Power conversion can operate momentarily without tripping any protection (100s of us)**
  - **Operation in this region is necessary when there is a fault in a rack far from the input of this rack.**
- **Transient 3 Operation**
  - **Surge/dropout condition (TBD)**
  - **Hold up time necessary to safely shut down the different converters in the system/tray to avoid damage.**

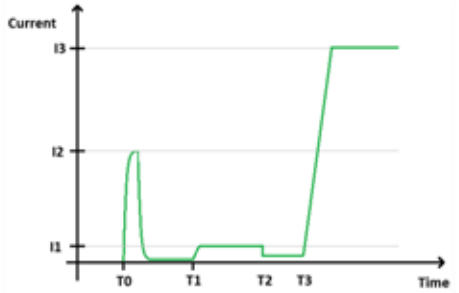
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Joint Rack&Power workstream

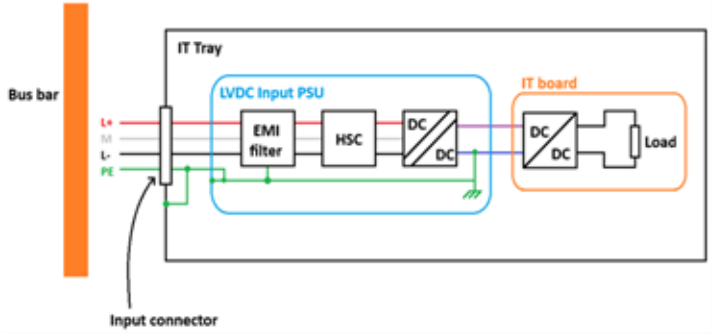
LVDC Input Power Supply Architecture and Grounding



Generalized LVDC input PSU architecture



Input Current During Different Stages



Example grounding system in an IT tray and a bipolar LVDC input PSU

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Control & Operation

- Sequence Of Operation (SOO)
  - SOO in AC Versus DC and the requirements
- The candidates for Interoperability and Communications Protocols
- Importance of Stability Study

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Control & Operation

Domain	Likely protocol / interface class	SOO role
MV switchgear, relays, protection IEDs	IEC 61850, GOOSE, MMS, sampled values where applicable	Fast protection signaling, breaker status, interlocks, event records
Utility / substation SCADA	DNP3 / IEEE 1815, IEC 61850, utility-specified interface	PCC telemetry, utility ride-through, trip/reclose, demand response
Plant controllers, PLCs, EPMS/BMS	OPC UA, Modbus RTU/TCP, BACnet/IP, vendor APIs	Supervisory control, monitoring, alarms, setpoints
MV DC UPS, SST and other Rectifiers, DC/DC for Battery Back Up, BESS PCS, SSCBs.	Digital Interface Protocol for Electronic Instrument Transformer, Real-time Industrial Ethernet Fieldbus Protocol and vendor defined proprietary protocols	System state machine, fault management, Interlock, Operation & Maintenance, System re-energization,
Time synchronization	IEEE 1588 PTP / IEC 61850-9-3 where required	Time-aligned sequence-of-events and fault analysis

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EMC

- Outlining existing Standards for Emissions and Immunity looking at both NA and Global applications
- Proposed testing for AC/DC, AC/DC/DC
- Outlining test levels, grounding, shielding, and filtering considerations
- Applicable loading and testing of power converters and systems
- Future needs need to consider EMC tests and levels for DC/DC converters as well as cooling distribution units and IT racks.

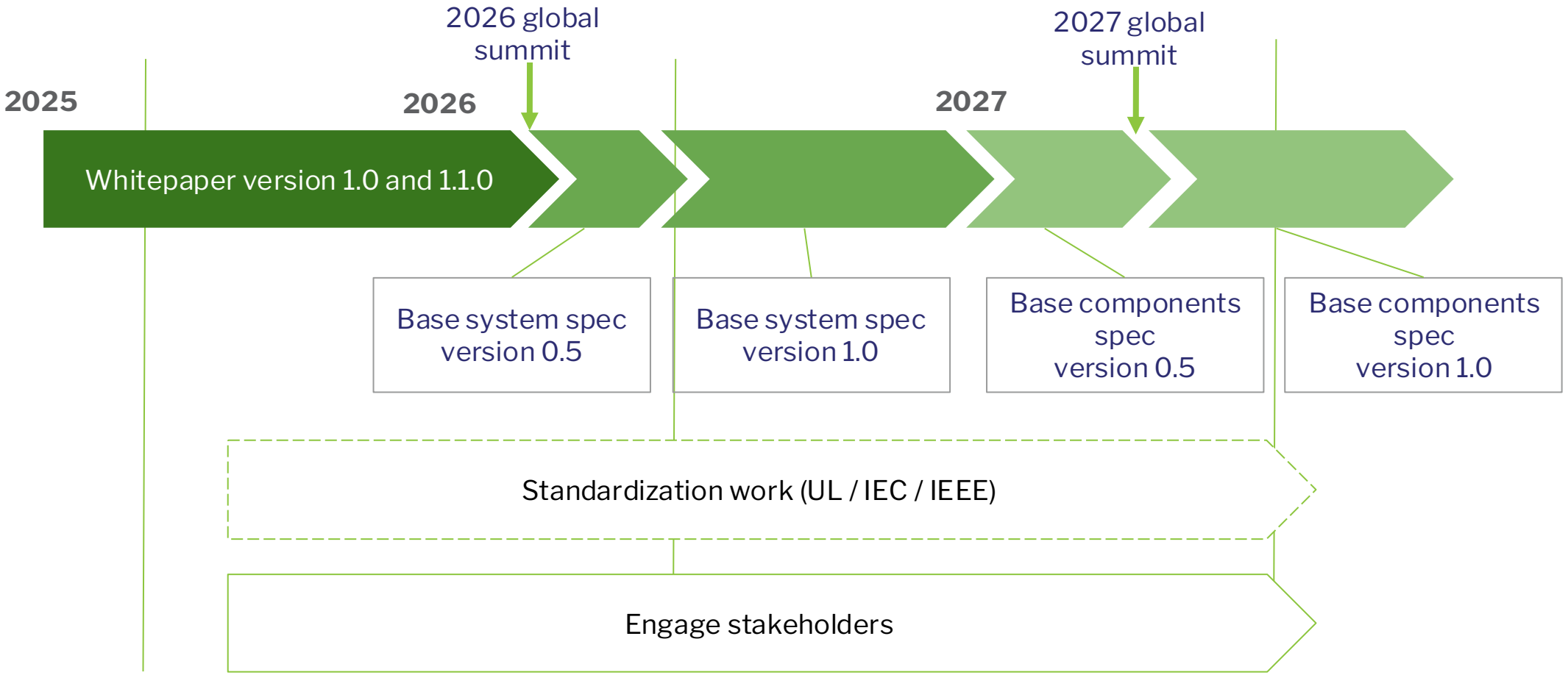
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Codes&Standards

- Updated Standard inventory
- Highlighted critical areas

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What's next (still work in progress)



Call to Action

- Share your comments to white paper Version 1.1.0
- Join the sub-project community
- Contribute to future work on base specs
- Where to find additional information (URL links)
  - Sub-project page with calendar Power Distribution » Open Compute Project
  - Sub-Project Wiki page Data Center Facility/Power – OpenCompute

Thank You!

